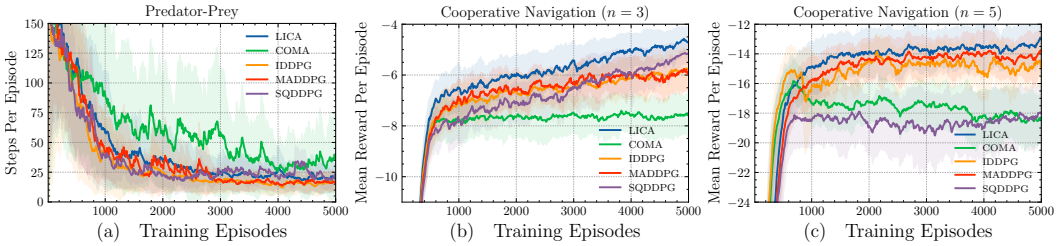

Figure 3: Comparing performance during training in **Multi-Agent Particle Environments**. (a) Predator-Prey. (b-c) Cooperative Navigation with $n = 3$ and $n = 5$ respectively.

Results. Fig. 3 reports the results for the Particle Environments. We first observe that LICA performs on par with or better than previous state-of-the-art methods, verifying its effectiveness. We can also see that LICA consistently outperforms COMA and MADDPG, confirming its capacity for credit assignment. Moreover, $n = 5$ for Cooperative Navigation increases the task difficulty as SQDDPG now underperforms and MADDPG yields better relative performance possibly due to the growth of its complexity with n. For Predator-Prey, LICA achieves similar convergence speed and performance compared to other methods, suggesting a saturation with the environment likely due to the random (instead of learned) prey. Despite achieving competitive results, we believe that the particle environments may not be sufficiently challenging to validate our methods, given that IDDPG, which completely ignores cooperation and/or credit assignment, is already competitive.

4.2 StarCraft II Micromanagement

Configurations. StarCraft II (SC2) provides a rich set of heterogeneous units each with diverse actions, allowing extremely complex cooperative behaviors among agents. We thus further evaluate LICA on several SC2 micromanagement tasks from the SMAC [39] benchmark, where a group of mixed-typed units controlled by decentralized agents needs to cooperate to defeat another group of mixed-typed enemy units controlled by built-in heuristic rules with "difficult" setting; the battles can be both symmetric (same units in both groups) or asymmetric. Same as previous work [10, 37, 46, 4], we used the default environment settings from SMAC. Each agent observes its own status and, within its field of view, it also observes other units' statistics such as health, location, and unit type; agents can only attack enemies within their shooting range. A shared reward is received on battle victory as well as damaging or killing enemy units. Each battle has step limits set by SMAC and may end early. We consider 6 battle maps grouped [39] into **Easy** (`2s3z`, `1c3s5z`), **Hard** (`5m_vs_6m`, `2c_vs_64zg`), and **Super Hard** (`MMM2`, `3s5z_vs_3s6z`) against 5 baseline methods using their open-source implementations based on PyMARL [39]: COMA [10], LIIR [4], VDN [46], QMIX [37], and QTRAN [45].

Training Settings. The inherent differences across the baseline methods and their training procedure (e.g. on/off-policy learning for policy/value-based methods) make it difficult to juxtapose them without introducing extra components (e.g. importance sampling [29, 50, 5] for off-policy policy evaluation) that could alter the baseline performance. To this end, [39] scales down both the batch size and the number of batch updates for on-policy methods in an effort to align their sample efficiency against off-policy methods with experience replay; while reasonable, this makes it hard to attribute any poor performance of on-policy methods to either the poor training conditions (particularly high variance from small batch sizes and insufficient gradient steps) or the underlying algorithmic limitations. To focus on the latter, we instead follow [4] where all methods use batches of 32 episodes generated with parallel runners to align the number of both batch updates and environment steps, which is in total 32 million steps for Easy (due to fast convergence) and 64 million for Hard and Super Hard scenarios. All methods also use GRU modules (Fig. 1) and share the parameters of the individual agent networks. Performance is evaluated every 320k env steps with 32 test episodes, and we report the median, 1st, and 3rd quartile win rates across five random seeds for every method in every map. See Supplementary for further environment and training details and battle visualizations.

Results. Fig. 4 reports the results across all 6 scenarios.[3] We first observe that LICA achieves competitive win rates across homogeneous and heterogeneous groups (e.g. `5m_vs_6m` and `2s3z`),

[3]Code is available at https://github.com/mzho7212/LICA. At the time of writing, our experiments are based on the latest PyMARL framework which uses SC2.4.10 while the original results reported in [39] uses SC2.4.6. As indicated by the original authors, performance is not always comparable across SC2 versions.

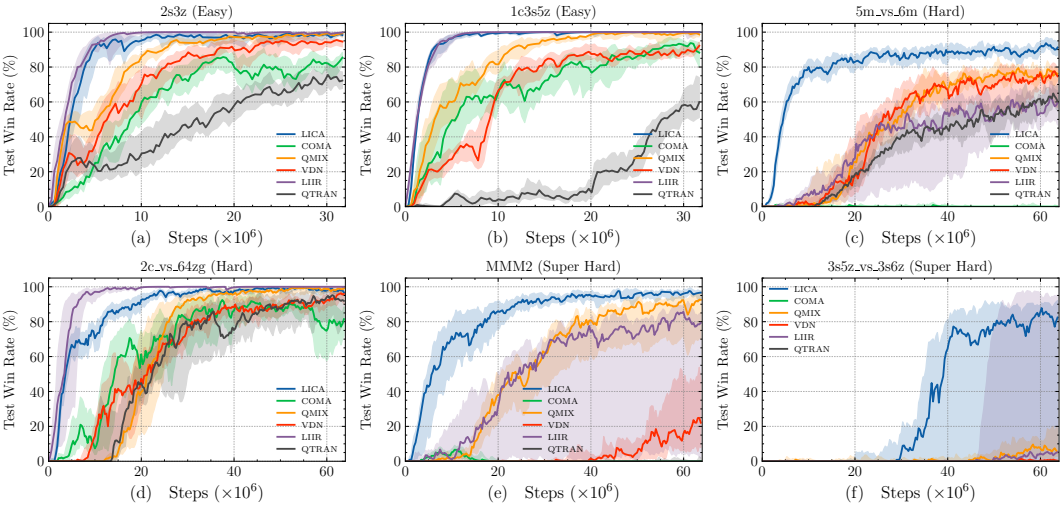

Figure 4: Performance comparison across various scenarios for **StarCraft II micromanagement**.

symmetric and asymmetric battles (e.g. `1c3s3z` and `3s5z_vs_3s6z`), and varying number of units (e.g. 2 units in `2c_vs_64zg` and 10 units in `MMM2`), underpinning its robustness. LICA also tends to give more consistent results than other methods across different difficulties. All methods solve Easy scenarios reasonably well, though the increased joint action diversity of `1c3s5z` over `2s5z` slows down convergence for certain methods while LICA is unaffected due to its high capacity. For the asymmetric `5m_vs_6m` (Hard), basic agent coordination alone such as "focus firing" [10, 37, 39, 4] no longer suffices and consistent success requires extended exploration to uncover complex cooperative strategies such as pulling back units with low health during combat. In `2c_vs_64zg`, while LICA outperforms most methods and reaches nearly perfect win rate, it underperforms LIIR in convergence speed; this is possibly because an optimal control of only 2 Colossus units may prioritize individual performance over cooperation where LIIR could benefit from its explicit formulation of an individual reward. Nevertheless, on the most challenging scenarios `MMM2` (involving 10 units of 3 types) and `3s5z_vs_3s6z` (where the extra enemy Zealot gives significant advantage), LICA's considerable performance improvements confirm the effectiveness of our proposed methods.

4.3 Component Studies

4.3.1 1-Step Traffic Junction

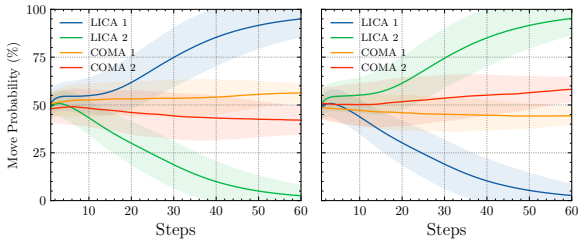

Figure 5: 1-Step Traffic Junction agent `move` probabilities throughout training for the two optimal outcomes.

To verify the credit assignment capacity of LICA, we introduce a simple 1-step environment where multiple optimal joint actions exist. The motivation here is to use a minimal scenario to illustrate the limited capacity of explicit credit assignment methods based on difference rewards for learning optimal policies, and having multiple optima is one such scenario that is also common in the real world (e.g. traffic management). The 1-step game involves two vehicles controlled by two separate agents attempting to pass a traffic junction in order. Each agent can either `pass` or `wait`; if both try to `pass` or `wait` at the same time, they receive a shared reward of 0; otherwise they receive a shared reward of 1, resulting in two optimal joint actions. It is worth noting that the simplicity of this 1-step game obviates most key aspects that differentiate on/off-policy learning (e.g. replay buffers and separate target/behavior nets) and focuses only on the mechanism for credit assignment. Taking COMA [10] as a representative example, difference reward based agents will expect a counterfactual advantage of 0 over the four possible joint actions; that is, if the centralized critic captures the true joint action value function, then on average, COMA agents perform as good as random agents. In practice, however, imperfect value functions and/or stochastic gradient descent (as opposed to full-batch gradient descent to consider all possible counterfactual actions) do allow agents to converge to optimal policies. To experiment, we train LICA and COMA agents for 20k random initializations each with 60 steps and

show the mean and standard deviation of the agent action probabilities for the two optimal outcomes in Fig. 5. The fast policy convergence and low policy entropy of LICA verifies the effectiveness of its implicit credit assignment.

4.3.2 Ablation Experiments

We further perform ablation experiments on SC2 `5m_vs_6m` (Hard) to validate each proposed component in LICA.

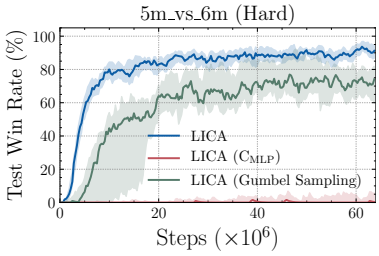

Figure 6: Ablations for different LICA components on SC2 `5m_vs_6m`.

Mixing Critic. We first consider replacing the centralized mixing critic (C_{Mix}) with an MLP critic (C_{MLP}, Section 3.2) which maps the concatenated representations of the state and actions into the joint Q estimate. In Fig. 2 (b-c), we perform two runs (labeled with C_{MLP}) against the best LICA run ($\xi = 0.11$) with different entropy regularization strengths, where $\xi = 0.11$ (C_{MLP}) provides a controlled comparison and $\xi = 0.05$ (C_{MLP}) leads to a similar entropy level. We also provide repeated runs with $\xi = 0.05$ (C_{MLP}) in Fig. 6. The poor performance with C_{MLP} in all experiments (see also Supplementary) clearly indicates the necessity of the mixing critic.

Policy Optimization with Action Distribution Parameters. We next consider how policy learning using action distribution parameters (i.e. probabilistic actions, Eq. 4) instead of sampled actions may affect performance. In Fig. 6, we report the results of LICA agents trained with "Straight-Through" Gumbel-Softmax [20] on `5m_vs_6m` (results on more SC2 maps are deferred to Supplementary). Intriguingly, despite the critic may not provide good estimates for probabilistic actions, we observe that LICA policies trained with Eq. 4 often converge faster and more stably to better optima than those learned with Gumbel sampling (though we believe the latter can improve its stability with well-crafted temperature schedules). One useful intuition is that action distribution parameters are more informative about the agents' behaviors than sampled actions [52], and as "softer" (yet deterministic to the agent observations) inputs to the critic, they could lead to less variance in the resulting value gradients [16, 20]; from a learning perspective, both of these properties can be favorable to policy optimization. Despite the empirical evidence supporting this argument, however, it would be interesting to develop further theoretical insights into this training regime in future work.

Adaptive Entropy Regularization. We further perform an ablation study to validate the proposed adaptive entropy regularization scheme and report the results in Fig. 2. We first observe that the adaptive control of regularization strength (with ξ) leads to sustained levels of policy stochasticity throughout training, which is not the case with vanilla entropy regularization (with β). This property also makes tuning regularization strengths easier: increasing the strength for LICA from the sub-optimal $\xi = 0.06$ where the policy converges prematurely to the optimal $\xi = 0.11$ leads to a *predictable* change in the entropy trajectory, which is also the case for LICA (C_{MLP}) when ξ is raised from 0.05 to 0.11; in contrast, when tuning the default entropy regularization ($\beta \in \{0.13, 0.15, 0.17\}$), slight changes in initial strength can lead to either similar or vastly different entropy trajectories. We also see that while $\beta = 0.17$ can reach similar win rates as $\xi = 0.11$, the agent exploration levels are drastically different before and after converging to exploitable policies (i.e. leap in win rates); in contrast, adaptive regularization encourages a consistent stochasticity level where the policies improve steadily and converge faster. We believe that given a large β and sufficient training, the vanilla formulation can indeed lead to optimal policies due to LICA's inherent high capacity, though the empirical results verify the benefits of the proposed adaptive entropy regularization.

5 Conclusions and Future Work

This paper presents LICA, a new actor-critic method that aims to implicitly address the credit assignment problem for fully cooperative multi-agent RL problems. LICA learns the decentralized policies by directly ascending the approximate joint action value gradients, and it makes use of two key components: a centralized critic hypernetwork that integrates a state representation into the policy gradients, and an adaptive entropy regularization scheme to dynamically control the policy stochasticity for encouraging extended and consistent agent exploration throughout training. Compared to previous methods, LICA has a simpler and a more general formulation, has higher capacity for credit assignment, and performs competitively in practice. For future work, we aim to improve its sample-efficiency by integrating off-policy training techniques; extend it for multi-agent continuous control; and explore further theoretical properties of adaptive entropy regularization.